PART VIII

Groups in Topology†

Section 41 Simplicial Complexes and Homology Groups
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SECTION 41 Simplicial Complexes and Homology Groups

Motivation

Topology concerns sets for which we have enough of an idea of when two points are close together to be able to define a continuous function. Two such sets, or *topological spaces,* are structurally the same if there is a one-to-one function mapping one onto the other such that both this function and its inverse are continuous. Naively, this means that one space can be stretched, twisted, and otherwise deformed, without being torn or cut, to look just like the other. Thus a big sphere is topologically the same structure as a small sphere, the boundary of a circle the same structure as the boundary of a square, and so on. Two spaces that are structurally the same in this sense are **homeomorphic.** Hopefully the student recognizes that *the concept of homeomorphism is to topology as the concept of isomorphism (where sets have the same algebraic structure) is to algebra.*

The *main problem of topology* is to find useful, necessary and sufficient conditions, other than just the definition, for two spaces to be homeomorphic. Sufficient conditions are hard to come by in general. Necessary conditions are a dime a dozen, but some are very important and useful. A "nice" space has associated with it various kinds of groups, namely *homology groups, cohomology groups, homotopy groups,* and *cohomotopy groups.* If two spaces are homeomorphic, it can be shown that the groups of one are isomorphic to the corresponding groups associated with the other. Thus a necessary condition for spaces to be homeomorphic is that their groups be isomorphic. Some of these groups may reflect very interesting properties of the spaces. Moreover, a continuous mapping of one space into another gives rise to homomorphisms from the groups

† Part VIII is not required for the remainder of the text.

of one into the groups of the other. These group homomorphisms may reflect interesting properties of the mapping.

If the student could make neither head nor tail out of the preceding paragraphs, he need not worry. The above paragraphs were just intended as motivation for what follows. It is the purpose of this section to describe some groups, *homology groups*, that are associated with certain simple spaces, in our work, usually some subset of the familiar Euclidean 3-space $\mathbb{R}^3$.

Preliminary Notions

First we introduce the idea of an *oriented n-simplex* in Euclidean 3-space $\mathbb{R}^3$ for $n = 0, 1, 2,$ and 3. An **oriented 0-simplex** is just a point P. An **oriented 1-simplex** is a *directed* line segment P_1P_2 joining the points P_1 and P_2 and viewed as traveled in the direction from P_1 to P_2. Thus $P_1P_2 \neq P_2P_1$. We will agree, however, that $P_1P_2 = -P_2P_1$. An **oriented 2-simplex** is a triangular region $P_1P_2P_3$, as in Fig. 41.1, together with a prescribed order of movement around the triangle, e.g., indicated by the arrow in Fig. 41.1 as the order $P_1P_2P_3$. The order $P_1P_2P_3$ is clearly the same order as $P_2P_3P_1$ and $P_3P_1P_2$, but the opposite order from $P_1P_3P_2$, $P_3P_2P_1$ and $P_2P_1P_3$. We will agree that

$$P_1P_2P_3 = P_2P_3P_1 = P_3P_1P_2 = -P_1P_3P_2 = -P_3P_2P_1 = -P_2P_1P_3.$$

Note that $P_iP_jP_k$ is equal to $P_1P_2P_3$ if

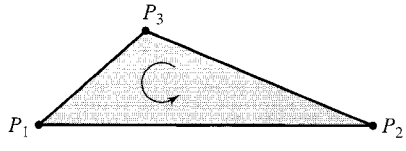
$$\begin{pmatrix} 1 & 2 & 3 \\ i & j & k \end{pmatrix}$$

is an even permutation, and is equal to $-P_1P_2P_3$ if the permutation is odd. The same could be said for an oriented 1-simplex P_1P_2. Note also that for $n = 0, 1, 2$, an oriented n-simplex is an n-dimensional object.

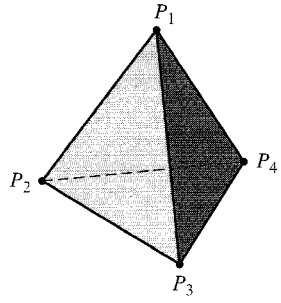
The definition of an oriented 3-simplex should now be clear: An **oriented 3-simplex** is given by an ordered sequence $P_1P_2P_3P_4$ of four vertices of a solid tetrahedron, as in Fig. 41.2. We agree that $P_1P_2P_3P_4 = \pm P_iP_jP_rP_s$, depending on whether the permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ i & j & r & s \end{pmatrix}$$

is even or odd. Similar definitions hold for $n > 3$, but we shall stop here with dimensions that we can visualize. These simplexes are **oriented,** or have an **orientation,** meaning that we are concerned with the *order* of the vertices as well as with the actual points where the vertices are located. All our simplexes will be oriented, and we shall drop the adjective from now on.



41.1 Figure



41.2 Figure

We are now going to define the *boundary of an n-simplex* for $n = 0, 1, 2, 3$. The term *boundary* is intuitive. We define the **boundary of a 0-simplex** P to be the empty simplex, which we denote this time by "0." The notation is

$$\text{“}\partial_0(P) = 0.\text{”}$$

The **boundary of a 1-simplex** P_1P_2 is defined by

$$\partial_1(P_1P_2) = P_2 - P_1,$$

that is, the formal difference of the end point and the beginning point. Likewise, the **boundary of a 2-simplex** is defined by

$$\partial_2(P_1P_2P_3) = P_2P_3 - P_1P_3 + P_1P_2,$$

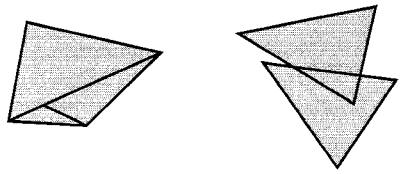
which we can remember by saying that it is the formal sum of terms that we obtain by dropping each P_i in succession from the 2-simplex $P_1P_2P_3$ and taking the sign to be $+$ if the first term is omitted, $-$ if the second is omitted, and $+$ if the third is omitted. Referring to Fig. 41.1, we see that this corresponds to going around what we naturally would call the boundary in the direction indicated by the orientation arrow. Note also that the equation $\partial_1(P_1P_2) = P_2 - P_1$ can be remembered in the same way. Thus we are led to the following definition of the **boundary of a 3-simplex:**

$$\partial_3(P_1P_2P_3P_4) = P_2P_3P_4 - P_1P_3P_4 + P_1P_2P_4 - P_1P_2P_3.$$

Similar definitions hold for the definition of ∂_n for $n > 3$. Each individual *summand* of the boundary of a simplex is a **face of the simplex.** Thus, $P_2P_3P_4$ is a face of $P_1P_2P_3P_4$, but $P_1P_3P_4$ is not a face. However, $P_1P_4P_3 = -P_1P_3P_4$ is a face of $P_1P_2P_3P_4$.

Suppose that you have a subset of $\mathbb{R}^3$ that is divided up "nicely" into simplexes, as, for example, the *surface* S of the tetrahedron in Fig. 41.2, which is split up into four 2-simplexes nicely fitted together. Thus on the surface of the tetrahedron, we have some 0-simplexes, or the vertices, of the tetrahedron; some 1-simplexes, or the edges of the tetrahedron; and some 2-simplexes, or the triangles of the tetrahedron. In general, for a space to be divided up "nicely" into simplexes, we require that the following be true:

1. Each point of the space belongs to at least one simplex.
2. Each point of the space belongs to only a finite number of simplexes.



41.3 Figure

3. Two different (up to orientation) simplexes either have no points in common or one is (except possibly for orientation) a face of the other or a face of a face of the other, etc., or the set of points in common is (except possibly for orientation) a face, or a face of a face, etc., of each simplex.

Condition (3) excludes configurations like those shown in Fig. 41.3. A space divided up into simplexes according to these requirements is a **simplicial complex.**

Chains, Cycles, and Boundaries

Let us now describe some groups associated with a simplicial complex X. We shall illustrate each definition with the case of the *surface* S of our tetrahedron in Fig. 41.2. The **group $C_n(X)$ of (oriented) n-chains of X** is the free abelian group generated by the (oriented) n-simplexes of X. Thus every element of $C_n(X)$ is a finite sum of the form $\sum_i m_i\sigma_i$, where the σ_i are n-simplexes of X and $m_i \in \mathbf{Z}$. We accomplish addition of chains by taking the algebraic sum of the coefficients of each occurrence in the chains of a fixed simplex.

41.4 Example For the surface S of our tetrahedron, every element of $C_2(S)$ is of the form

$$m_1P_2P_3P_4 + m_2P_1P_3P_4 + m_3P_1P_2P_4 + m_4P_1P_2P_3$$

for $m_i \in \mathbf{Z}$. As an illustration of addition, note that

$$(3P_2P_3P_4 - 5P_1P_2P_3) + (6P_2P_3P_4 - 4P_1P_3P_4) = 9P_2P_3P_4 - 4P_1P_3P_4 - 5P_1P_2P_3.$$

An element of $C_1(S)$ is of the form

$$m_1P_1P_2 + m_2P_1P_3 + m_3P_1P_4 + m_4P_2P_3 + m_5P_2P_4 + m_6P_3P_4,$$

and an element of $C_0(S)$ is of the form

$$m_1P_1 + m_2P_2 + m_3P_3 + m_4P_4. \quad \blacktriangle$$

Now if σ is an n-simplex, $\partial_n(\sigma) \in C_{n-1}(X)$ for $n = 1, 2, 3$. Let us define $C_{-1}(X) = \{0\}$, the trivial group of one element, and then we will also have $\partial_0(\sigma) \in C_{-1}(X)$. Since $C_n(X)$ is free abelian, and since we can specify a homomorphism of such a group by giving its values on generators, we see that ∂_n gives a unique **boundary homomorphism,** which we denote again by "∂_n," mapping $C_n(X)$ into $C_{n-1}(X)$ for $n = 0, 1, 2, 3$.

41.5 Example We have

$$\partial_n\left(\sum_i m_i\sigma_i\right) = \sum_i m_i\partial_n(\sigma_i).$$

For example,

$$\begin{aligned}\partial_1(3P_1P_2 - 4P_1P_3 + 5P_2P_4) &= 3\partial_1(P_1P_2) - 4\partial_1(P_1P_3) + 5\partial_1(P_2P_4)\\ &= 3(P_2 - P_1) - 4(P_3 - P_1) + 5(P_4 - P_2)\\ &= P_1 - 2P_2 - 4P_3 + 5P_4.\end{aligned}$$

▲

The student is reminded again that *any time you have a homomorphism, two things are of great interest, the kernel and the image.* The kernel of ∂_n consists of those n-chains with boundary 0. The elements of the kernel are ***n*-cycles.** The usual notation for the kernel of ∂_n, that is, the **group of *n*-cycles,** is "$Z_n(X)$."

41.6 Example If $z = P_1P_2 + P_2P_3 + P_3P_1$, then

$$\partial_1(z) = (P_2 - P_1) + (P_3 - P_2) + (P_1 - P_3) = 0.$$

Thus z is a 1-cycle. However, if we let $c = P_1P_2 + 2P_2P_3 + P_3P_1$, then

$$\partial_1(c) = (P_2 - P_1) + 2(P_3 - P_2) + (P_1 - P_3) = -P_2 + P_3 \neq 0.$$

Thus $c \notin Z_1(X)$. ▲

Note that $z = P_1P_2 + P_2P_3 + P_3P_1$ of Example 41.6 corresponds to one circuit, or *cycle*, around a triangle with vertices P_1, P_2, and P_3.

The image under ∂_n, the **group of $(n-1)$-boundaries,** consists exactly of those $(n-1)$-chains that are boundaries of n-chains. This group is denoted by "$B_{n-1}(X)$."

41.7 Example Referring to Example 41.6, we see that if

$$P_1P_2 + 2P_2P_3 + P_3P_1$$

is a 1-chain in $C_1(X)$, then $P_3 - P_2$ is a 0-boundary. Note that $P_3 - P_2$ *bounds* P_2P_3. ▲

Let us now compute $Z_n(X)$ and $B_n(X)$ for a more complicated example. In topology, if a group is the trivial group consisting just of the identity 0, one usually denotes it by "0" rather than "$\{0\}$." We shall follow this convention.

41.8 Example Let us compute for $n = 0, 1, 2$ the groups $Z_n(S)$ and $B_n(S)$ for the *surface* S of the tetrahedron of Fig. 41.2.

First, for the easier cases, since the highest dimensional simplex for the surface is a 2-simplex, we have $C_3(S) = 0$, so

$$B_2(S) = \partial_3[C_3(S)] = 0.$$

Also, since $C_{-1}(S) = 0$ by our definition, we see that

$$Z_0(S) = C_0(S).$$

Thus $Z_0(S)$ is free abelian on four generators, P_1, P_2, P_3, and P_4. It is easily seen that the image of a group under a homomorphism is generated by the images of generators of the original group. Thus, since $C_1(S)$ is generated by P_1P_2, P_1P_3, P_1P_4, P_2P_3, P_2P_4, and P_3P_4, we see that $B_0(S)$ is generated by

$$P_2 - P_1,\ P_3 - P_1,\ P_4 - P_1,\ P_3 - P_2,\ P_4 - P_2,\ P_4 - P_3.$$

However, $B_0(S)$ is not free abelian on these generators. For example, $P_3 - P_2 = (P_3 - P_1) - (P_2 - P_1)$. It is easy to see that $B_0(S)$ is free abelian on $P_2 - P_1$, $P_3 - P_1$, and $P_4 - P_1$.

Now let us go after the tougher group $Z_1(S)$. An element c of $C_1(S)$ is a formal sum of integral multiples of edges P_iP_j. It is clear that $\partial_1(c) = 0$ if and only if each vertex that is the beginning point of a total (counting multiplicity) of r edges of c is also the end point of exactly r edges. Thus

$$\begin{aligned} z_1 &= P_2P_3 + P_3P_4 + P_4P_2, \\ z_2 &= P_1P_4 + P_4P_3 + P_3P_1, \\ z_3 &= P_1P_2 + P_2P_4 + P_4P_1, \\ z_4 &= P_1P_3 + P_3P_2 + P_2P_1 \end{aligned}$$

are all 1-cycles. These are exactly the boundaries of the individual 2-simplexes. We claim that the z_i generate $Z_1(S)$. Let $z \in Z_1(S)$, and choose a particular vertex, say P_1; let us work on edges having P_1 as an end point. These edges are P_1P_2, P_1P_3, and P_1P_4. Let the coefficient of P_1P_j in z be m_j. Then

$$z + m_2z_4 - m_4z_2$$

is again a cycle, but does not contain the edges P_1P_2 or P_1P_4. Thus the only edge having P_1 as a vertex in the cycle $z + m_2z_4 - m_4z_2$ is possibly P_1P_3, but this edge could not appear with a nonzero coefficient as it would contribute a nonzero multiple of the vertex P_1 to the boundary, contradicting the fact that a cycle has boundary 0. Thus $z + m_2z_4 - m_4z_2$ consists of the edges of the 2-simplex $P_2P_3P_4$. Since in a 1-cycle each of P_2, P_3, and P_4 must serve the same number of times as a beginning and an end point of edges in the cycle, counting multiplicity, we see that

$$z + m_2z_4 - m_4z_2 = rz_1$$

for some integer r. Thus $Z_1(S)$ is generated by the z_i, actually by any three of the z_i. Since the z_i are the individual boundaries of the 2-simplexes, as we observed, we see that

$$Z_1(S) = B_1(S).$$

The student should see geometrically what this computation means in terms of Fig. 41.2.

Finally, we describe $Z_2(S)$. Now $C_2(S)$ is generated by the simplexes $P_2P_3P_4$, $P_3P_1P_4$, $P_1P_2P_4$, and $P_2P_1P_3$. If $P_2P_3P_4$ has coefficient r_1 and $P_3P_1P_4$ has coefficient r_2 in a 2-cycle, then the common edge P_3P_4 has coefficient $r_1 - r_2$ in its boundary.

Thus we must have $r_1 = r_2$, and by a similar argument, in a cycle each one of the four 2-simplexes appears with the same coefficient. Thus $Z_2(S)$ is generated by

$$P_2P_3P_4 + P_3P_1P_4 + P_1P_2P_4 + P_2P_1P_3,$$

that is, $Z_2(S)$ is infinite cyclic. Again, the student should interpret this computation geometrically in terms of Fig. 41.2. Note that the orientation of each summand corresponds to going around that triangle clockwise, when viewed from the outside of the tetrahedron. ▲

$\partial^2 = 0$ and Homology Groups

We now come to one of the most important equations in all of mathematics. We shall state it only for $n = 1, 2$, and 3, but it holds for all $n > 0$.

41.9 Theorem Let X be a simplicial complex, and let $C_n(X)$ be the n-chains of X for $n = 0, 1, 2, 3$. Then the composite homomorphism $\partial_{n-1}\partial_n$ mapping $C_n(X)$ into $C_{n-2}(X)$ maps everything into 0 for $n = 1, 2, 3$. That is, for each $c \in C_n(X)$ we have $\partial_{n-1}(\partial_n(c)) = 0$. We use the notation "$\partial_{n-1}\partial_n = 0$," or, more briefly, "$\partial^2 = 0$."

Proof Since a homomorphism is completely determined by its values on generators, it is enough to check that for an n-simplex σ, we have $\partial_{n-1}(\partial_n(\sigma)) = 0$. For $n = 1$ this is obvious, since ∂_0 maps everything into 0. For $n = 2$,

$$\begin{aligned}\partial_1(\partial_2(P_1P_2P_3)) &= \partial_1(P_2P_3 - P_1P_3 + P_1P_2)\\ &= (P_3 - P_2) - (P_3 - P_1) + (P_2 - P_1)\\ &= 0.\end{aligned}$$

The case $n = 3$ will make an excellent exercise for the student in the definition of the boundary operator (see Exercise 2). ◆

41.10 Corollary For $n = 0$, 1, 2, and 3, $B_n(X)$ is a subgroup of $Z_n(X)$.

Proof For $n = 0$, 1, and 2, we have $B_n(X) = \partial_{n+1}[C_{n+1}(X)]$. Then if $b \in B_n(X)$, we must have $b = \partial_{n+1}(c)$ for some $c \in C_{n+1}(X)$. Thus

$$\partial_n(b) = \partial_n(\partial_{n+1}(c)) = 0,$$

so $b \in Z_n(X)$.

For $n = 3$, since we are not concerned with simplexes of dimension greater than 3, $B_3(X) = 0$. ◆

41.11 Definition The factor group $H_n(X) = Z_n(X)/B_n(X)$ is the ***n*-dimensional homology group of *X***. ■

41.12 Example Let us calculate $H_n(S)$ for $n = 0$, 1, 2, and 3 and where S is the *surface* of the tetrahedron in Fig. 41.2.

We found $Z_n(S)$ and $B_n(S)$ in Example 41.8. Now $C_3(S) = 0$, so $Z_3(S)$ and $B_3(S)$ are both 0, and hence

$$H_3(S) = 0.$$

Also, $Z_2(S)$ is infinite cyclic and we saw that $B_2(S) = 0$. Thus $H_2(S)$ is infinite cyclic, that is,

$$H_2(S) \simeq \mathbb{Z}.$$

We saw that $Z_1(S) = B_1(S)$, so the factor group $Z_1(S)/B_1(S)$ is the trivial group of one element, that is,

$$H_1(S) = 0.$$

Finally, $Z_0(S)$ was free abelian on P_1, P_2, P_3, and P_4, while $B_0(S)$ was generated by $P_2 - P_1$, $P_3 - P_1$, $P_4 - P_1$, $P_3 - P_2$, $P_4 - P_2$, and $P_4 - P_3$. We claim that every coset of $Z_0(S)/B_0(S)$ contains exactly one element of the form rP_1. Let $z \in Z_0(S)$, and suppose that the coefficient of P_2 in z is s_2, of P_3 is s_3, and of P_4 is s_4. Then

$$z - [s_2(P_2 - P_1) + s_3(P_3 - P_1) + s_4(P_4 - P_1)] = rP_1$$

for some r, so $z \in [rP_1 + B_0(S)]$, that is, any coset does contain an element of the form rP_1. If the coset also contains $r'P_1$, then $r'P_1 \in [rP_1 + B_0(S)]$, so $(r' - r)P_1$ is in $B_0(S)$. Clearly, the only multiple of P_1 that is a boundary is zero, so $r = r'$ and the coset contains exactly one element of the form rP_1. We may then choose the rP_1 as representatives of the cosets in computing $H_0(S)$. Thus $H_0(S)$ is infinite cyclic, that is,

$$H_0(S) \simeq \mathbb{Z}. \qquad \blacktriangle$$

These definitions and computations probably seem very complicated to the student. The ideas are very natural, but we admit that they are a bit messy to write down. However, the arguments used in these calculations are typical for homology theory, i.e., if you can understand them, you will understand all our others. Furthermore, we can make them *geometrically*, looking at the picture of the space. The next section will be devoted to further computations of homology groups of certain simple but important spaces.

EXERCISES 41

Suggested Exercises

1. Assume that $c = 2P_1P_3P_4 - 4P_3P_4P_6 + 3P_3P_2P_4 + P_1P_6P_4$ is a 2-chain of a certain simplicial complex X.
   a. Compute $\partial_2(c)$. **b.** Is c a 2-cycle? **c.** Is $\partial_2(c)$ a 1-cycle?
2. Compute $\partial_2(\partial_3(P_1P_2P_3P_4))$ and show that it is 0, completing the proof of Theorem 41.9.
3. Describe $C_i(P)$, $Z_i(P)$, $B_i(P)$, and $H_i(P)$ for the space consisting of just the 0-simplex P. (This is really a trivial problem.)
4. Describe $C_i(X)$, $Z_i(X)$, $B_i(X)$, and $H_i(X)$ for the space X consisting of two distinct 0-simplexes, P and P'. (*Note:* The line segment joining the two points is *not* part of the space.)
5. Describe $C_i(X)$, $Z_i(X)$, $B_i(X)$, and $H_i(X)$ for the space X consisting of the 1-simplex P_1P_2.
6. Mark each of the following true or false.
   _____ **a.** Every boundary is a cycle.
   _____ **b.** Every cycle is a boundary.
   _____ **c.** $C_n(X)$ is always a free abelian group.

_____ **d.** $B_n(X)$ is always a free abelian group.
_____ **e.** $Z_n(X)$ is always a free abelian group.
_____ **f.** $H_n(X)$ is always abelian.
_____ **g.** The boundary of a 3-simplex is a 2-simplex.
_____ **h.** The boundary of a 2-simplex is a 1-chain.
_____ **i.** The boundary of a 3-cycle is a 2-chain.
_____ **j.** If $Z_n(X) = B_n(X)$, then $H_n(X)$ is the trivial group of one element.

More Exercises

7. Define the following concepts so as to generalize naturally the definitions in the text given for dimensions 0, 1, 2, and 3.
   a. An oriented n-simplex
   b. The boundary of an oriented n-simplex
   c. A face of an oriented n-simplex

8. Continuing the idea of Exercise 7, what would be an easy way to answer a question asking you to define $C_n(X)$, $\partial_n : C_n(X) \to C_{n-1}(X)$, $Z_n(X)$, and $B_n(X)$ for a simplicial complex X perhaps containing some simplexes of dimension greater than 3?

9. Following the ideas of Exercises 7 and 8, prove that $\partial^2 = 0$ in general, i.e., that $\partial_{n-1}(\partial_n(c)) = 0$ for every $c \in C_n(X)$, where n may be greater than 3.

10. Let X be a simplicial complex. For an (oriented) n-simplex σ of X, the **coboundary $\delta^{(n)}(\sigma)$ of** σ is the $(n+1)$-chain $\sum \tau$, where the sum is taken over all $(n+1)$-simplexes τ that have σ as a face. That is, the simplexes τ appearing in the sum are precisely those that have σ as a *summand* of $\partial_{n+1}(\tau)$. *Orientation is important here.* Thus P_2 is a face of P_1P_2, but P_1 is not. However, P_1 is a face of P_2P_1. Let X be the simplicial complex consisting of the *solid tetrahedron* of Fig. 41.2.
   a. Compute $\delta^{(0)}(P_1)$ and $\delta^{(0)}(P_4)$.
   b. Compute $\delta^{(1)}(P_3P_2)$.
   c. Compute $\delta^{(2)}(P_3P_2P_4)$.

11. Following the idea of Exercise 10, let X be a simplicial complex, and let the group $C^{(n)}(X)$ of ***n*-cochains** be the same as the group $C_n(X)$.
   a. Define $\delta^{(n)} : C^{(n)}(X) \to C^{(n+1)}(X)$ in a way analogous to the way we defined $\partial_n : C_n(X) \to C_{n-1}(X)$.
   b. Show that $\delta^2 = 0$, that is, that $\delta^{(n+1)}(\delta^{(n)}(c)) = 0$ for each $c \in C^{(n)}(X)$.

12. Following the ideas of Exercises 10 and 11, define the *group $Z^{(n)}(X)$ of n-cocycles* of X, the *group $B^{(n)}(X)$ of n-coboundaries* of X, and show that $B^{(n)}(X) \leq Z^{(n)}(X)$.

13. Following the ideas of Exercises 10, 11, and 12, define the *n-dimensional cohomology group* $H^{(n)}(X)$ of X. Compute $H^{(n)}(S)$ for the *surface* S of the tetrahedron of Fig. 41.2.

SECTION 42 Computations of Homology Groups

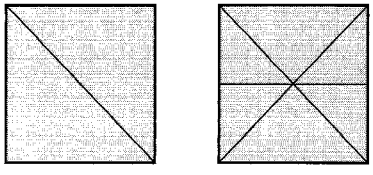
Triangulations

Suppose you wish to calculate homology groups for the surface of a sphere. The first thing you probably will say, if you are alert, is that the surface of a sphere is not a simplicial complex, since this surface is curved and a triangle is a plane surface. Remember that two spaces are topologically the same if one can be obtained from the other by bending,

twisting, and so on. Imagine our 3-simplex, the tetrahedron, to have a rubber surface and to be filled with air. If the rubber surface is flexible, like the rubber of a balloon, it will promptly deform itself into a sphere and the four faces of the tetrahedron will then appear as "triangles" drawn on the surface of the sphere. This illustrates what is meant by a *triangulation* of a space. The term *triangulation* need not refer to a division into 2-simplexes only, but is also used for a division into n-simplexes for any $n \geq 0$. If a space is divided up into pieces in such a way that near each point the space can be deformed to look like a part of some Euclidean space $\mathbb{R}^n$ and the pieces into which the space was divided appear after this deformation as part of a simplicial complex, then the original division of the space is a **triangulation of the space.** The homology groups of the space are then defined formally just as in the last section.

Invariance Properties

There are two very important *invariance properties* of homology groups, the proofs of which require quite a lot of machinery, but that are easy for us to explain roughly. First, the homology groups of a space are defined in terms of a triangulation, but actually they are the same (i.e., isomorphic) groups no matter how the space is triangulated. For example, a square region can be triangulated in many ways, two of which are shown in Fig. 42.1. The homology groups are the same no matter which triangulation is used to compute them. *This is not obvious!*



42.1 Figure

For the second invariance property, if one triangulated space is homeomorphic to another (e.g., can be deformed into the other without being torn or cut), the homology groups of the two spaces are the same (i.e., isomorphic) in each dimension n. *This is, again, not obvious.* We shall use both of these facts without proof.

42.2 Example The homology groups of the surface of a sphere are the same as those for the surface of our tetrahedron in Example 41.12, since the two spaces are homeomorphic. ▲

Two important types of spaces in topology are the spheres and the cells. Let us introduce them and the usual notations. The ***n*-sphere** S^n is the set of all points a distance of 1 unit from the origin in $(n+1)$-dimensional Euclidean space $\mathbb{R}^{n+1}$. Thus the 2-sphere S^2 is what is usually called the *surface* of a sphere in $\mathbb{R}^3$, S^1 is the rim of a circle, and S^0 is two points. Of course, the choice of 1 for the distance from the origin is not important. A 2-sphere of radius 10 is homeomorphic to one of radius 1 and homeomorphic to the surface of an ellipsoid for that matter. The ***n*-cell** or ***n*-ball** E^n is the set of all points in $\mathbb{R}^n$ a distance ≤ 1 from the origin. Thus E^3 is what you usually think of as a solid sphere, E^2 is a circular region, and E^1 is a line segment.

42.3 Example The above remarks and the computations of Example 41.12 show that $H_2(S^2)$ and $H_0(S^2)$ are both isomorphic to $\mathbb{Z}$, and $H_1(S^2) = 0$. ▲

Connected and Contractible Spaces

There is a very nice interpretation of $H_0(X)$ for a space X with a triangulation. A space is **connected** if any two points in it can be joined by a path (a concept that we will not define) lying totally in the space. If a space is not connected, then it is split up into a number of pieces, each of which is connected but no two of which can be joined by a path in the space. These pieces are the **connected components of the space.**

42.4 Theorem If a space X is triangulated into a finite number of simplexes, then $H_0(X)$ is isomorphic to $\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$, and the Betti number m of factors $\mathbb{Z}$ is the number of connected components of X.

Proof Now $C_0(X)$ is the free abelian group generated by the finite number of vertices P_i in the triangulation of X. Also, $B_0(X)$ is generated by expressions of the form

$$P_{i_2} - P_{i_1},$$

where $P_{i_1} P_{i_2}$ is an edge in the triangulation. Fix P_{i_1}. Any vertex P_{i_r} in the same connected component of X as P_{i_1} can be joined to P_{i_1} by a finite sequence

$$P_{i_1} P_{i_2}, P_{i_2} P_{i_3}, \ldots, P_{i_{r-1}} P_{i_r}$$

of edges. Then

$$P_{i_r} = P_{i_1} + (P_{i_2} - P_{i_1}) + (P_{i_3} - P_{i_2}) + \cdots + (P_{i_r} - P_{i_{r-1}}),$$

showing that $P_{i_r} \in [P_{i_1} + B_0(X)]$. Clearly, if P_{i_s} is not in the same connected component with P_{i_1}, then $P_{i_s} \notin [P_{i_1} + B_0(X)]$, since no edge joins the two components. Thus, if we select one vertex from each connected component, each coset of $H_0(X)$ contains exactly one representative that is an integral multiple of one of the selected vertices. The theorem follows at once. ◆

42.5 Example We have at once that

$$H_0(S^n) \simeq \mathbb{Z}$$

for $n > 0$, since S^n is connected for $n > 0$. However,

$$H_0(S^0) \simeq \mathbb{Z} \times \mathbb{Z}$$

(see Chapter 41, Exercise 4). Also,

$$H_0(E^n) \simeq \mathbb{Z}$$

for $n \geq 1$. ▲

A space is **contractible** if it can be compressed to a point without being torn or cut, *but always kept within the space it originally occupied.* We just state the next theorem.

42.6 Theorem If X is a contractible space triangulated into a finite number of simplexes, then $H_n(X) = 0$ for $n \geq 1$.

42.7 Example It is a fact that S^2 is not contractible. It is not too easy to prove this fact. The student will, however, probably be willing to take it as self-evident that you cannot compress the "surface of a sphere" to a point without tearing it, *keeping it always within the original space S^2 that it occupied.* It is not fair to compress it all to the "center of the sphere." We saw that $H_2(S^2) \neq 0$ but is isomorphic to $\mathbb{Z}$.

Suppose, however, we consider $H_2(E^3)$, where we can regard E^3 as our solid tetrahedron of Fig. 41.2, for it is homomorphic to E^3. The surface S of this tetrahedron is homomorphic to S^2. The simplexes here for E^3 are the same as they are for S (or S^2), which we examined in Examples 41.8 and 41.12, except for the whole 3-simplex σ that now appears. Remember that a generator of $Z_2(S)$, and hence of $Z_2(E^3)$, was exactly the entire boundary of σ. Viewed in E^3, this is $\partial_3(\sigma)$, an element of $B_2(E^3)$, so $Z_2(E^3) = B_2(E^3)$ and $H_2(E^3) = 0$. Since E^3 is obviously contractible, this is consistent with Theorem 42.6. ▲

In general, E^n is contractible for $n \geq 1$, so we have by Theorem 42.6,

$$H_i(E^n) = 0$$

for $i > 0$.

Further Computations

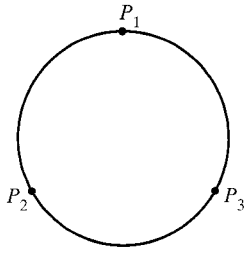
We have seen a nice interpretation for $H_0(X)$ in Theorem 42.4. As the preceding examples illustrate, the 1-cycles in a triangulated space are generated by closed curves of the space formed by edges of the triangulation. The 2-cycles can be thought of as generated by 2-spheres or other closed 2-dimensional surfaces in the space. Forming the factor group

$$H_1(X) = Z_1(X)/B_1(X)$$

amounts roughly to counting the closed curves that appear in the space that are not there simply because they appear as the boundary of a 2-dimensional piece (i.e., a collection of 2-simplexes) of the space. Similarly, forming $H_2(X) = Z_2(X)/B_2(X)$ amounts roughly to counting the closed 2-dimensional surfaces in the space that cannot be "filled in solid" within the space, i.e., are not boundaries of some collection of 3-simplexes. Thus for $H_1(S^2)$, every closed curve drawn on the surface of the 2-sphere bounds a 2-dimensional piece of the sphere, so $H_1(S^2) = 0$. However, the only possible closed 2-dimensional surface, S^2 itself, cannot be "filled in solid" *within the whole space S^2 itself*, so $H_2(S^2)$ is free abelian on one generator.

42.8 Example According to the reasoning above, one would expect $H_1(S^1)$ to be free abelian on one generator, i.e., isomorphic to $\mathbb{Z}$, since the circle itself is not the boundary of a 2-dimensional part of S^1. You see, there *is* no 2-dimensional part of S^1. We compute and see whether this is indeed so.

A triangulation of S^1 is given in Fig. 42.9. Now $C_1(S^1)$ is generated by P_1P_2, P_2P_3, and P_3P_1. If a 1-chain is a cycle so that its boundary is zero, then it must contain P_1P_2



42.9 Figure

and P_2P_3 the same number of times, otherwise its boundary would contain a nonzero multiple of P_2. A similar argument holds for any two edges. Thus $Z_1(S^1)$ is generated by $P_1P_2 + P_2P_3 + P_3P_1$. Since $B_1(S^1) = \partial_2[C_2(S^1)] = 0$, there being no 2-simplexes, we see that $H_1(S^1)$ is free abelian on one generator, that is,

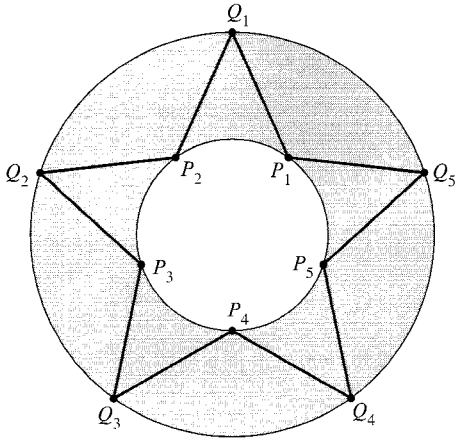
$$H_1(S^1) \simeq \mathbb{Z}. \qquad \blacktriangle$$

It can be proved that for $n > 0$, $H_n(S^n)$ and $H_0(S^n)$ are isomorphic to $\mathbb{Z}$, while $H_i(S^n) = 0$ for $0 < i < n$.

To conform to topological terminology, we shall call an element of $H_n(X)$, that is, a coset of $B_n(X)$ in $Z_n(X)$, a **"homology class."** Cycles in the same homology class are **homologous.**

42.10 Example Let us compute the homology groups of a plane annular region X between two concentric circles. A triangulation is indicated in Fig. 42.11. Of course, since X is connected, it follows that

$$H_0(X) \simeq \mathbb{Z}.$$



42.11 Figure

If z is any 1-cycle, and if P_1P_2 has coefficient r in z, then $z - r\partial_2(P_1P_2Q_1)$ is a cycle without P_1P_2 homologous to z. By continuing this argument, we find that there is a 1-cycle homologous to z containing no edge on the inner circle of the annulus. Using the "outside" triangles, we can adjust further by multiples of $\partial_2(Q_iP_iQ_j)$, and we arrive at z' containing no edge Q_iP_i either. But then if Q_5P_1 appears in z' with nonzero coefficient, P_1 appears with nonzero coefficient in $\partial_1(z')$, contradicting the fact that z' is a cycle. Similarly, no edge Q_iP_{i+1} can occur for $i = 1, 2, 3, 4$. Thus z is homologous to a cycle made up of edges only on the outer circle. By a familiar argument, such a cycle must be of the form

$$n(Q_1Q_2 + Q_2Q_3 + Q_3Q_4 + Q_4Q_5 + Q_5Q_1).$$

It is then clear that

$$H_1(X) \simeq \mathbb{Z}.$$

We showed that we could "push" any 1-cycle to the outer circle. Of course, we could have pushed it to the inner circle equally well.

For $H_2(X)$, note that $Z_2(X) = 0$, since every 2-simplex has in its boundary an edge on either the inner or the outer circle of the annulus that appears in no other 2-simplex. The boundary of any nonzero 2-chain must then contain some nonzero multiples of these edges. Hence

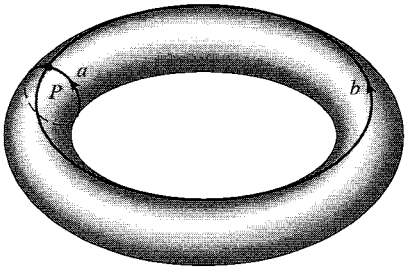
$$H_2(X) = 0.$$

▲

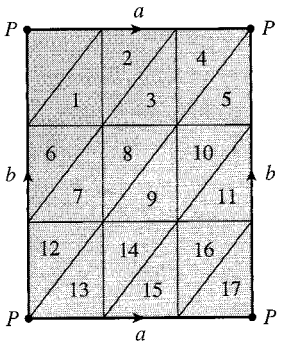
42.12 Example We shall compute the homology groups of the torus surface X which looks like the surface of a doughnut, as in Fig. 42.13. To visualize a triangulation of the torus, imagine that you cut it on the circle marked a, then cut it all around the circle marked b, and flatten it out as in Fig. 42.14. Then draw the triangles. To recover the torus from Fig. 42.14, join the left edge b to the right edge b in such a way that the arrows are going in the same direction. This gives a cylinder with circle a at each end. Then bend the cylinder around and join the circles a, again keeping the arrows going the same way around the circles.

Since the torus is connected, $H_0(X) \simeq \mathbb{Z}$.

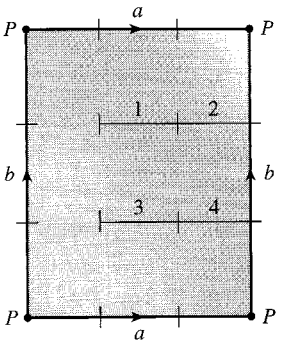
For $H_1(X)$, let z be a 1-cycle. By changing z by a multiple of the boundary of the triangle numbered 1 in Fig. 42.14, you can get a homologous cycle not containing the side / of triangle 1. Then by changing this new 1-cycle by a suitable multiple of the boundary of triangle 2, you can further eliminate the side | of 2. Continuing, we can then



42.13 Figure



42.14 Figure



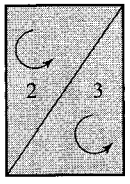
42.15 Figure

eliminate / of 3, | of 4, / of 5, — of 6, / of 7, | of 8, / of 9, | of 10, / of 11, — of 12, / of 13, | of 14, / of 15, | of 16, and / of 17. The resulting cycle, homologous to z, can then only contain the edges shown in Fig. 42.15. But such a cycle could not contain, with nonzero coefficient, any of the edges we have numbered in Fig. 42.15, or it would not have boundary 0. Thus z is homologous to a 1-cycle having edges only on the circle a or the circle b (refer to Fig. 42.13). By a now hopefully familiar argument, every edge on circle a must appear the same number of times, and the same is also true for edges on circle b; however, an edge on circle b need not appear the same number of times as an edge appears on a. Furthermore, if a 2-chain is to have a boundary just containing a and b, all the triangles oriented counterclockwise must appear with the same coefficient so that the inner edges will cancel out. The boundary of such a 2-chain is 0. Thus every homology class (coset) contains one and only one element

$$ra + sb,$$

where r and s are integers. Hence $H_1(X)$ is free abelian on two generators, represented by the two circles a and b. Therefore,

$$H_1(X) \simeq \mathbb{Z} \times \mathbb{Z}.$$



42.16 Figure

Finally, for $H_2(X)$, a 2-cycle must contain the triangle numbered 2 of Fig. 42.14 with counterclockwise orientation the same number of times as it contains the triangle numbered 3, also with counterclockwise orientation, in order for the common edge / of these triangles not to be in the boundary. These orientations are illustrated in Fig. 42.16. The same holds true for any two adjacent triangles, and thus every triangle with the counterclockwise orientation must appear the same number of times in a 2-cycle. Clearly, any multiple of the formal sum of all the 2-simplexes, all with counterclockwise orientation, is a 2-cycle. Thus $Z_2(X)$ is infinite cyclic, isomorphic to $\mathbb{Z}$. Also, $B_2(X) = 0$, there being no 3-simplexes, so

$$H_2(X) \simeq \mathbb{Z}.$$

▲

EXERCISES 42

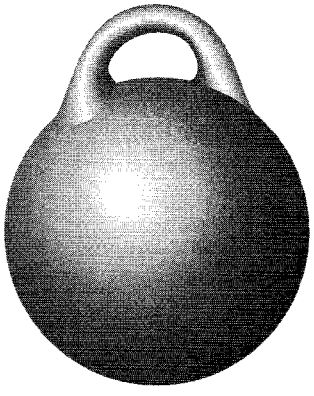
In these exercises, you need not write out in detail your computations or arguments.

Computations

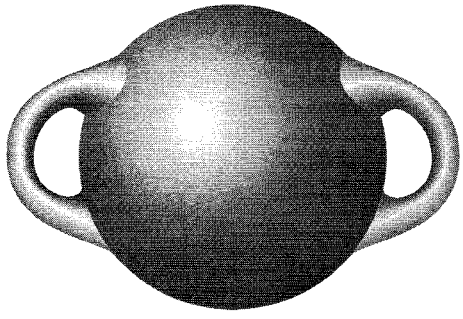
1. Compute the homology groups of the space consisting of two tangent 1-spheres, i.e., a figure eight.
2. Compute the homology groups of the space consisting of two tangent 2-spheres.
3. Compute the homology groups of the space consisting of a 2-sphere with an annular ring (as in Fig. 42.11) that does not touch the 2-sphere.
4. Compute the homology groups of the space consisting of a 2-sphere with an annular ring whose inner circle is a great circle of the 2-sphere.
5. Compute the homology groups of the space consisting of a circle touching a 2-sphere at one point.
6. Compute the homology groups of the surface consisting of a 2-sphere with a handle (see Fig. 42.17).
7. Mark each of the following true or false.

   _____ **a.** Homeomorphic simplicial complexes have isomorphic homology groups.
   _____ **b.** If two simplicial complexes have isomorphic homology groups, then the spaces are homeomorphic.
   _____ **c.** S^n is homeomorphic to E^n.
   _____ **d.** $H_n(X)$ is trivial for $n > 0$ if X is a connected space with a finite triangulation.
   _____ **e.** $H_n(X)$ is trivial for $n > 0$ if X is a contractible space with a finite triangulation.
   _____ **f.** $H_n(S^n) = 0$ for $n > 0$.
   _____ **g.** $H_n(E^n) = 0$ for $n > 0$.
   _____ **h.** A torus is homeomorphic to S^2.
   _____ **i.** A torus is homeomorphic to E^2.
   _____ **j.** A torus is homeomorphic to a sphere with a handle on it (see Fig. 42.17).

8. Compute the homology groups of the space consisting of two torus surfaces having no points in common.



42.17 Figure



42.18 Figure

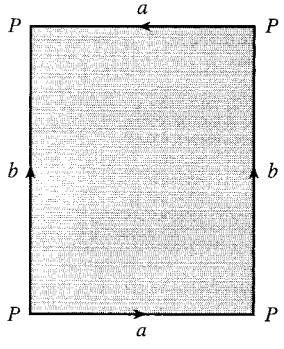
9. Compute the homology groups of the space consisting of two stacked torus surfaces, stacked as one would stack two inner tubes.
10. Compute the homology groups of the space consisting of a torus tangent to a 2-sphere at all points of a great circle of the 2-sphere, i.e., a balloon wearing an inner tube.
11. Compute the homology groups of the surface consisting of a 2-sphere with two handles (see Fig. 42.18).
12. Compute the homology groups of the surface consisting of a 2-sphere with n handles (generalizing Exercises 6 and 11.

SECTION 43 More Homology Computations and Applications

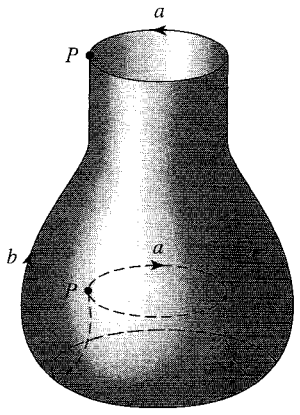
One-Sided Surfaces

Thus far all the homology groups we have found have been free abelian, so that there were no nonzero elements of finite order. This can be shown always to be the case for the homology groups of a *closed surface* (i.e., a surface like S^2, which has no boundary) that has two sides. Our next example is of a *one-sided closed surface,* the *Klein bottle*. Here the 1-dimensional homology group will have a nontrivial torsion subgroup reflecting the twist in the surface.

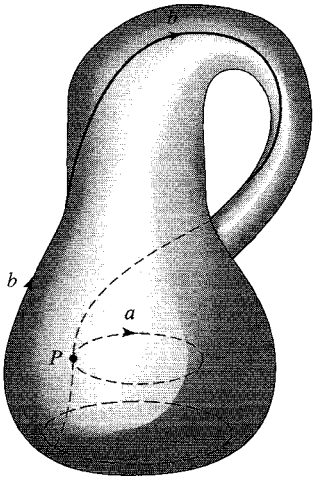
43.1 Example Let us calculate the homology groups of the Klein bottle X. Figure 43.2 represents the Klein bottle cut apart, just as Fig. 42.14 represents the torus cut apart. The only difference is that the arrows on the top and bottom edge a of the rectangle are in *opposite* directions this time. To recover a Klein bottle from Fig. 43.2, again bend the rectangle joining the edges labeled b so that the directions of the arrows match up. This gives a cylinder that is shown somewhat deformed, with the bottom end pushed a little way up inside the cylinder, in Fig. 43.3. Such deformations are legitimate in topology. Now the circles a must be joined so that the arrows go around the same way. *This cannot be done in* $\mathbb{R}^3$. You must imagine that you are in $\mathbb{R}^4$, so that you can bend the neck of the bottle around and "through" the side without intersecting the side, as shown in Fig. 43.4. With a little thought, you can see that this resulting surface really has only one side. That is, if you



43.2 Figure



43.3 Figure



43.4 Figure

start at any place and begin to paint "one side," you will wind up painting the whole thing. There is no concept of an *inside* of a Klein bottle.

We can calculate the homology groups of the Klein bottle much as we calculated the homology groups of the torus in Example 42.12, by splitting Fig. 43.2 into triangles exactly as we did for the torus. Of course,

$$H_0(X) \simeq \mathbb{Z},$$

since X is connected. As we found for the torus, if we triangulate the Klein bottle by dividing Fig. 43.2 into triangles, every 1-cycle is homologous to a cycle of the form

$$ra + sb$$

for r and s integers. If a 2-chain is to have a boundary containing just a and b, again, all the triangles oriented counterclockwise must appear with the same coefficient so that the inner edges will cancel each other. In the case of the torus, the boundary of such a 2-chain was 0. Here, however, it is $k(2a)$, where k is the number of times each triangle appears. Thus $H_1(X)$ is an abelian group with generators the homology classes of a and b and the relations $a + b = b + a$ and $2a = 0$. Therefore,

$$H_1(X) \simeq \mathbb{Z}_2 \times \mathbb{Z},$$

a group with torsion coefficient 2 and Betti number 1. Our argument above regarding 2-chains shows that there are no 2-cycles this time, so

$$H_2(X) = 0. \qquad \blacktriangle$$

A torsion coefficient does not have to be present in some homology group of a one-sided surface *with boundary*. Mostly for the sake of completeness, we give this standard example of the *Möbius strip*.

43.5 Example Let X be the Möbius strip, which we can form by taking a rectangle of paper and joining the two ends marked a with a half twist so that the arrows match up, as indicated in Fig. 43.6. Note that the Möbius strip is a surface with a boundary, and the boundary is just one closed curve (homomorphic to a circle) made up of l and l'. It is clear that the Möbius strip, like the Klein bottle, has just one side, in the sense that if you were asked to color only one side of it, you would wind up coloring the whole thing.

Of course, since X is connected,

$$H_0(X) \simeq \mathbb{Z}.$$

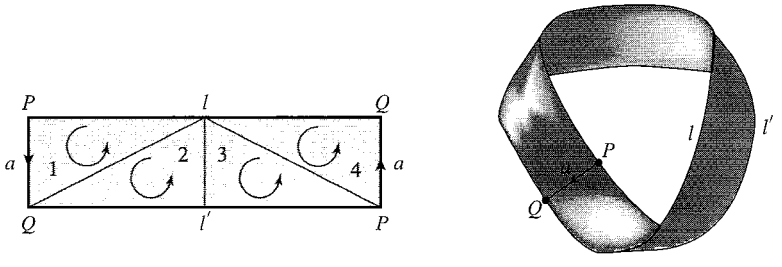
Let z be any 1-cycle. By subtracting in succession suitable multiples of the triangles numbered 2, 3, and 4 in Fig. 43.6, we can eliminate edges/of triangle 2, | of triangle 3, and \ of triangle 4. Thus z is homologous to a cycle z' having edges on only l, l', and a, and as before, both edges on l' must appear the same number of times. But if c is a 2-chain consisting of the formal sum of the triangles oriented as shown in Fig. 43.6, we see that $\partial_2(c)$ consists of the edges on l and l' plus $2a$. Since both edges on l' must appear in z' the same number of times, by subtracting a suitable multiple of $\partial_2(c)$, we see that z is homologous to a cycle with edges just lying on l and a. By a familiar argument, all these edges properly oriented must appear the same number of times in this new cycle, and thus the homology class containing their formal sum is a generator for $H_1(X)$. Therefore,

$$H_1(X) \simeq \mathbb{Z}.$$

This generating cycle starts at Q and goes around the strip, then cuts across it at P via a, and arrives at its starting point.

If z'' were a 2-cycle, it would have to contain the triangles 1, 2, 3, and 4 of Fig. 43.6 an equal number r of times with the indicated orientation. But then $\partial_2(z'')$ would be $r(2a + l + l') \neq 0$. Thus $Z_2(X) = 0$, so

$$H_2(X) = 0. \qquad \blacktriangle$$



43.6 Figure

The Euler Characteristic

Let us turn from the computation of homology groups to a few interesting facts and applications. Let X be a finite simplicial complex (or triangulated space) consisting of simplexes of dimension 3 and less. Let n_0 be the total number of vertices in the triangulation, n_1 the number of edges, n_2 the number of 2-simplexes, and n_3 the number

of 3-simplexes. The number

$$n_0 - n_1 + n_2 - n_3 = \sum_{i=0}^{3} (-1)^i n_i$$

can be shown to be the same no matter how the space X is triangulated. This number is the **Euler characteristic** $\chi(X)$ of the space. We just state the following fascinating theorem.

43.7 Theorem Let X be a finite simplicial complex (or triangulated space) of dimension ≤ 3. Let $\chi(X)$ be the Euler characteristic of the space X, and let β_j be the Betti number of $H_j(X)$. Then

$$\chi(X) = \sum_{j=0}^{3} (-1)^j \beta_j.$$

This theorem holds also for X of dimension greater than 3, with the obvious extension of the definition of the Euler characteristic to dimension greater than 3.

43.8 Example Consider the solid tetrahedron E^3 of Fig. 41.2. Here $n_0 = 4$, $n_1 = 6$, $n_2 = 4$, and $n_3 = 1$, so

$$\chi(E^3) = 4 - 6 + 4 - 1 = 1.$$

Remember that we saw that $H_3(E^3) = H_2(E^3) = H_1(E^3) = 0$ and $H_0(E^3) \simeq \mathbb{Z}$. Thus $\beta_3 = \beta_2 = \beta_1 = 0$ and $\beta_0 = 1$, so

$$\sum_{j=0}^{3} (-1)^j \beta_j = 1 = \chi(E^3).$$

For the surface S^2 of the tetrahedron in Fig. 41.2, we have $n_0 = 4$, $n_1 = 6$, $n_2 = 4$, and $n_3 = 0$, so

$$\chi(S^2) = 4 - 6 + 4 = 2.$$

Also, $H_3(S^2) = H_1(S^2) = 0$, and $H_2(S^2)$ and $H_0(S^2)$ are both isomorphic to $\mathbb{Z}$. Thus $\beta_3 = \beta_1 = 0$ and $\beta_2 = \beta_0 = 1$, so

$$\sum_{j=0}^{3} (-1)^j \beta_j = 2 = \chi(S^2).$$

Finally, for S^1 in Fig. 42.9, $n_0 = 3$, $n_1 = 3$, and $n_2 = n_3 = 0$, so

$$\chi(S^1) = 3 - 3 = 0.$$

Here $H_1(S^1)$ and $H_0(S^1)$ are both isomorphic to $\mathbb{Z}$, and $H_3(S^1) = H_2(S^1) = 0$. Thus $\beta_0 = \beta_1 = 1$ and $\beta_2 = \beta_3 = 0$, giving

$$\sum_{j=0}^{3} (-1)^j \beta_j = 0 = \chi(S^1).$$

▲

Mappings of Spaces

A continuous function f mapping a space X into a space Y gives rise to a homomorphism f_{*n} mapping $H_n(X)$ into $H_n(Y)$ for $n \geq 0$. The demonstration of the existence of this homomorphism takes more machinery than we wish to develop here, but let us attempt to describe how these homomorphisms can be computed in certain cases. The following is true:

> If $z \in \mathbb{Z}_n(X)$, and if $f(z)$, regarded as the result of picking up z and setting it down in Y in the naively obvious way, should be an n-cycle in Y, then
>
> $$f_{*n}(z + B_n(X)) = f(z) + B_n(Y).$$
>
> That is, if z represents a homology class in $H_n(X)$ and $f(z)$ is an n-cycle in Y, then $f(z)$ represents the image homology class under f_{*n} of the homology class containing z.

Let us illustrate this and attempt to show just what we mean here by $f(z)$.

43.9 Example Consider the unit circle

$$S^1 = \{(x, y) \mid x^2 + y^2 = 1\}$$

in $\mathbb{R}^2$. Any point in S^1 has coordinates $(\cos\theta, \sin\theta)$, as indicated in Fig. 43.10. Let $f : S^1 \to S^1$ be given by

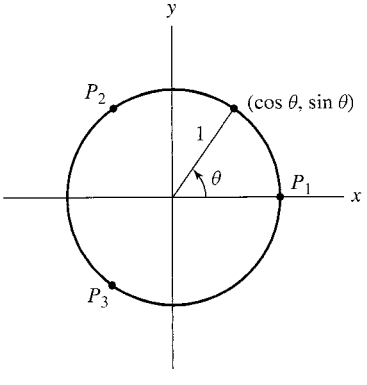
$$f((\cos\theta, \sin\theta)) = (\cos 3\theta, \sin 3\theta).$$

Obviously, this function f is continuous. Now f should induce

$$f_{*1} : H_1(S^1) \to H_1(S^1).$$

Here $H_1(S^1)$ is isomorphic to $\mathbb{Z}$ and has as generator the homology class of $z = P_1P_2 + P_2P_3 + P_3P_1$, as seen in Example 42.8. Now if P_1, P_2, and P_3 are evenly spaced about the circle, then f maps each of the arcs P_1P_2, P_2P_3, and P_3P_1 onto the whole perimeter of the circle, that is,

$$f(P_1P_2) = f(P_2P_3) = f(P_3P_1) = P_1P_2 + P_2P_3 + P_3P_1.$$



43.10 Figure

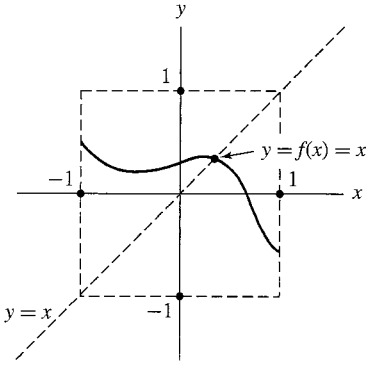
Thus

$$f_{*1}(z + B_1(S^1)) = 3(P_1P_2 + P_2P_3 + P_3P_1) + B_1(S^1)$$
$$= 3z + B_1(S^1),$$

that is, f_{*1} maps a generator of $H_1(S^1)$ onto three times itself. This obviously reflects the fact that f winds S^1 around itself three times. ▲

Example 43.9 illustrates our previous assertion that the homomorphisms of homology groups associated with a continuous mapping f may mirror important properties of the mapping.

Finally, we use these concepts to indicate a proof of the famous *Brouwer Fixed-Point Theorem*. This theorem states that a continuous map f of E^n into itself has a *fixed point*, i.e., there is some $x \in E^n$ such that $f(x) = x$. Let us see what this means for E^2, a circular region. Imagine that you have a thin sheet of rubber stretched out on a table to form a circular disk. Mark with a pencil the outside boundary of the rubber circle on the table. Then stretch, compress, bend, twist, and fold the rubber in any fashion without tearing it, but keep it always within the penciled circle on the table. When you finish, some point on the rubber will be over exactly the same point on the table at which it first started.

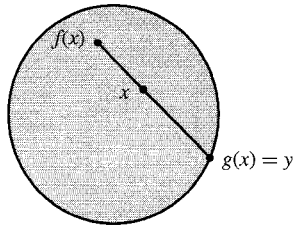


43.11 Figure

The proof we outline is good for any $n > 1$. For $n = 1$, looking at the graph of a function $f : E^1 \to E^1$, we find that the theorem simply states that any continuous path joining the left and right sides of a square must cross the diagonal somewhere, as indicated in Fig. 43.11. The student should visualize the construction of our proof with E^3 having boundary S^2 and E^2 having boundary S^1. The proof contains a figure illustrating the construction for the case of E^2.

43.12 Theorem (Brouwer Fixed-Point Theorem). A continuous map f of E^n into itself has a fixed point for $n \geq 1$.

Proof The case $n = 1$ was considered above. Let f be a map of E^n into E^n for $n > 1$. We shall assume that f has no fixed point and shall derive a contradiction.



43.13 Figure

If $f(x) \neq x$ for all $x \in E^n$, we can consider the line segment from $f(x)$ to x. Let us extend this line segment *in the direction from* $f(x)$ *to* x until it goes through the boundary S^{n-1} of E^n at some point y. This defines for us a function $g : E^n \to S^{n-1}$ with $g(x) = y$, as illustrated in Fig. 43.13. Note that for y on the boundary, we have $g(y) = y$. Now since f is continuous, it is pretty obvious that g is also continuous. (A **continuous function** is roughly one that maps points that are sufficiently close together into points that are close together. If x_1 and x_2 are sufficiently close together, then $f(x_1)$ and $f(x_2)$ are sufficiently close together so that the line segment joining $f(x_1)$ and x_1 is so close to the line segment joining $f(x_2)$ and x_2 that $y_1 = g(x_1)$ is close to $y_2 = g(x_2)$.) Then g is a continuous mapping of E^n into S^{n-1}, and thus induces a homomorphism

$$g_{*(n-1)} : H_{n-1}(E^n) \to H_{n-1}(S^{n-1}).$$

Now we said that $H_{n-1}(E^n) = 0$, for $n > 1$, since E^n is contractible, and we checked it for $n \neq 2$ and $n = 3$. Since $g_{*(n-1)}$ is a homomorphism, we must have $g_{*(n-1)}(0) = 0$. But an $(n-1)$-cycle representing the homology class 0 of $H_{n-1}(E^n)$ is the whole complex S^{n-1} with proper orientation of simplexes, and $g(S^{n-1}) = S^{n-1}$, since $g(y) = y$ for all $y \in S^{n-1}$. Thus

$$g_{*(n-1)}(0) = S^{n-1} + B_{n-1}(S^{n-1}).$$

which is a generator $\neq 0$ of $H_{n-1}(S^{n-1})$, a contradiction. ◆

We find the preceding proof very satisfying aesthetically, and hope you agree.

EXERCISES 43

Suggested Exercises

1. Verify by direct calculation that both triangulations of the square region X in Fig. 42.1 give the same value for the Euler characteristic $\chi(X)$.
2. Illustrate Theorem 43.7, as we did in Example 43.8, for each of the following spaces.
   - **a.** The annular region of Example 42.10
   - **b.** The torus of Example 42.12
   - **c.** The Klein bottle of Example 43.1

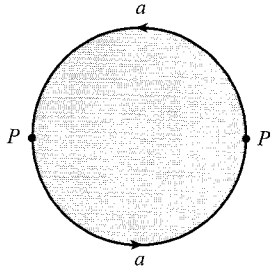
3. Will every continuous map of a square region into itself have a fixed point? Why or why not? Will every continuous map of a space consisting of two disjoint 2-cells into itself have a fixed point? Why or why not?
4. Compute the homology groups of the space consisting of a 2-sphere touching a Klein bottle at one point.
5. Compute the homology groups of the space consisting of two Klein bottles with no points in common.
6. Mark each of the following true or false.

   _____ **a.** Every homology group of a contractible space is the trivial group of one element.
   _____ **b.** A continuous map from a simplicial complex X into a simplicial complex Y induces a homomorphism of $H_n(X)$ into $H_n(Y)$.
   _____ **c.** All homology groups are abelian.
   _____ **d.** All homology groups are free abelian.
   _____ **e.** All 0-dimensional homology groups are free abelian.
   _____ **f.** If a space X has n-simplexes but none of dimension greater than n and $H_n(X) \neq 0$, then $H_n(X)$ is free abelian.
   _____ **g.** The boundary of an n-chain is an $(n-1)$-chain.
   _____ **h.** The boundary of an n-chain is an $(n-1)$-cycle.
   _____ **i.** The n-boundaries form a subgroup of the n-cycles.
   _____ **j.** The n-dimensional homology group of a simplicial complex is always a subgroup of the group of n-chains.

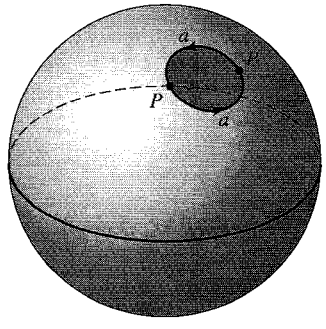
More Exercises

7. Find the Euler characteristic of a 2-sphere with n handles (see Section 42, Exercise 12).
8. We can form the topological *real projective plane* X, using Fig. 43.14, by joining the semicircles a so that diametrically opposite points come together and the directions of the arrows match up. *This cannot be done in Euclidean 3-space* $\mathbb{R}^3$. One must go to $\mathbb{R}^4$. Triangulate this space X, starting with the form exhibited in Fig. 43.14, and compute its homology groups.
9. The circular disk shown in Fig. 43.14 can be deformed topologically to appear as a 2-sphere with a hole in it, as shown in Fig. 43.15. We form the real projective plane from this configuration by sewing up the hole in such a way that only diametrically opposite points on the rim of the hole are sewn together. This cannot be done in $\mathbb{R}^3$.

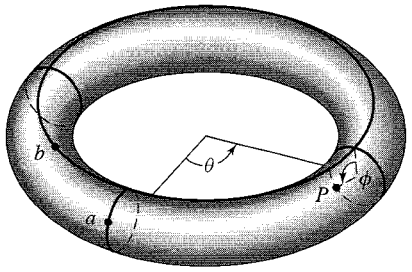
   Extending this idea, a 2-sphere with q holes in it, which are then sewn up by bringing together diametrically opposite points on the rims of the holes, gives a **2-sphere with q cross caps.** Find the homology groups of a 2-sphere with q cross caps. (To see a triangulation, view the space as the disk in Fig. 43.14 but with $q-1$ holes in it to be sewn up as described above. Then triangulate this disk with these holes.)



43.14 Figure



43.15 Figure



43.16 Figure

Comment: It can be shown that every sufficiently nice closed surface, namely a *closed 2-manifold,* is homeomorphic to a 2-sphere with some number $h \geq 0$ of handles if the surface is two sided, and is homeomorphic to a 2-sphere with $q > 0$ cross caps if it is one sided. The number h or q, as the case may be, is the **genus of the surface.**

10. Every point P on a regular torus X can be described by means of two angles θ and ϕ, as shown in Fig. 43.16. That is, we can associate *coordinates* (θ, ϕ) with P. For each of the mappings f of the torus X onto itself given below, describe the induced map f_{*n} of $H_n(X)$ into $H_n(X)$ for $n = 0$, 1, and 2, by finding the images of the generators for $H_n(X)$ described in Example 42.12. Interpret these group homomorphisms geometrically as we did in Example 43.9.

a. $f : X \to X$ given by $f((\theta, \phi)) = (2\theta, \phi)$
b. $f : X \to X$ given by $f((\theta, \phi)) = (\theta, 2\phi)$
c. $f : X \to X$ given by $f((\theta, \phi)) = (2\theta, 2\phi)$

11. With reference to Exercise 10, the torus X can be mapped onto its circle b (which is homeomorphic to S^1) by a variety of maps. For each such map $f : X \to b$ given below, describe the homomorphism $f_{*n} : H_n(X) \to H_n(b)$ for $n = 0$, 1, and 2, by describing the image of generators of $H_n(X)$ as in Exercise 10.

a. $f : X \to b$ given by $f((\theta, \phi)) = (\theta, 0)$
b. $f : X \to b$ given by $f((\theta, \phi)) = (2\theta, 0)$

12. Repeat Exercise 11, but view the map f as a map of the torus X into itself, inducing maps $f_{*n} : H_n(X) \to H_n(X)$.

13. Consider the map f of the Klein bottle in Fig. 43.2 given by mapping a point Q of the rectangle in Fig. 43.2 onto the point of b directly opposite (closest to) it. Note that b is topologically a 1-sphere. Compute the induced maps $f_{*n} : H_n(X) \to H_n(b)$ for $n = 0$, 1, and 2, by describing images of generators of $H_n(X)$.

SECTION 44 HOMOLOGICAL ALGEBRA

Chain Complexes and Mappings

The subject of algebraic topology was responsible for a surge in a new direction in algebra. You see, if you have a simplicial complex X, then you naturally get chain groups $C_k(X)$ and maps ∂_k, as indicated in the diagram

$$C_n(X) \xrightarrow{\partial_n} C_{n-1}(X) \xrightarrow{\partial_{n-1}} \cdots \xrightarrow{\partial_2} C_1(X) \xrightarrow{\partial_1} C_0(X) \xrightarrow{\partial_0} 0,$$

with $\partial_{k-1}\partial_k = 0$. You then abstract the purely algebraic portion of this situation and consider any sequence of abelian groups A_k and homomorphisms $\partial_k : A_k \to A_{k-1}$ such that $\partial_{k-1}\partial_k = 0$ for $k \geq 1$. So that you do not always have to require $k \geq 1$ in $\partial_{k-1}\partial_k = 0$, it is convenient to consider "doubly infinite" sequences of groups A_k for all $k \in \mathbb{Z}$. Often, $A_k = 0$ for $k < 0$ and $k > n$ in applications. The study of such sequences and maps of such sequences is a topic of *homological algebra.*

44.1 Definition **A chain complex** $\langle A, \partial \rangle$ is a doubly infinite sequence

$$A = \{\cdots, A_2, A_1, A_0, A_{-1}, A_{-2}, \cdots\}$$

of abelian groups A_k, together with a collection $\partial = \{\partial_k \mid k \in \mathbb{Z}\}$ of homomorphisms such that $\partial_k : A_k \to A_{k-1}$ and $\partial_{k-1}\partial_k = 0$. ■

As a convenience similar to our notation in group theory, we shall be sloppy and let "A" denote the chain complex $\langle A, \partial \rangle$. We can now imitate in a completely algebraic setting our constructions and definitions of Section 41.

44.2 Theorem If A is a chain complex, then the image under ∂_k is a subgroup of the kernel of ∂_{k-1}.

Proof Consider

$$A_k \xrightarrow{\partial_k} A_{k-1} \xrightarrow{\partial_{k-1}} A_{k-2}.$$

Now $\partial_{k-1}\partial_k = 0$, since A is a chain complex. That is, $\partial_{k-1}[\partial_k[A_k]] = 0$. This tells us at once that $\partial_k[A_k]$ is contained in the kernel of ∂_{k-1}, which is what we wished to prove. ◆

44.3 Definition If A is a chain complex, then the kernel $Z_k(A)$ of ∂_k is the **group of k-cycles,** and the image $B_k(A) = \partial_{k+1}[A_{k+1}]$ is the **group of k-boundaries.** The factor group $H_k(A) = Z_k(A)/B_k(A)$ is the **kth homology group of** A. ■

We stated in the last section that for simplicial complexes X and Y, a continuous mapping f from X into Y induces a homomorphism of $H_k(X)$ into $H_k(Y)$. This mapping of the homology groups arises in the following way. For suitable triangulations of X and Y, the mapping f gives rise to a homomorphism f_k of $C_k(X)$ into $C_k(Y)$, which has the important property that *it commutes with* ∂_k, that is,

$$\partial_k f_k = f_{k-1}\partial_k.$$

Let us turn to the purely algebraic situation and see how this induces a map of the homology groups.

44.4 Theorem **(Fundamental Lemma)** Let A and A' with collections ∂ and ∂' of homomorphisms be chain complexes, and suppose that there is a collection f of homomorphisms $f_k : A_k \to A'_k$ as indicated in the diagram

$$\begin{array}{ccccccccccc}
\cdots & \xrightarrow{\partial_{k+2}} & A_{k+1} & \xrightarrow{\partial_{k+1}} & A_k & \xrightarrow{\partial_k} & A_{k-1} & \xrightarrow{\partial_{k-1}} & \cdots \\
 & & \downarrow f_{k+1} & & \downarrow f_k & & \downarrow f_{k-1} & & \\
\cdots & \xrightarrow{\partial'_{k+2}} & A'_{k+1} & \xrightarrow{\partial'_{k+1}} & A'_k & \xrightarrow{\partial'_k} & A'_{k-1} & \xrightarrow{\partial'_{k-1}} & \cdots.
\end{array}$$

Suppose, furthermore, that every square is commutative, that is,

$$f_{k-1}\partial_k = \partial'_k f_k$$

for all k. Then f_k induces a natural homomorphism $f_{*k} : H_k(A) \to H_k(A')$.

Proof Let $z \in Z_k(A)$. Now

$$\partial'_k(f_k(z)) = f_{k-1}(\partial_k(z)) = f_{k-1}(0) = 0,$$

so $f_k(z) \in Z_k(A')$. Let us attempt to define $f_{*k} : H_k(A) \to H_k(A')$ by

$$f_{*k}(z + B_k(A)) = f_k(z) + B_k(A'). \tag{1}$$

We must first show that f_{*k} is well defined, i.e., independent of our choice of a representative of $z + B_k(A)$. Suppose that $z_1 \in (z + B_k(A))$. Then $(z_1 - z) \in B_k(A)$, so there exists $c \in A_{k+1}$ such that $z_1 - z = \partial_{k+1}(c)$. But then

$$f_k(z_1) - f_k(z) = f_k(z_1 - z) = f_k(\partial_{k+1}(c)) = \partial'_{k+1}(f_{k+1}(c)),$$

and this last term is an element of $\partial'_{k+1}[A'_{k+1}] = B_k(A')$. Hence

$$f_k(z_1) \in (f_k(z) + B_k(A')).$$

Thus two representatives of the same coset in $H_k(A) = Z_k(A)/B_k(A)$ are mapped into representatives of just one coset in $H_k(A') = Z_k(A')/B_k(A')$. This shows that $f_{*k} : H_k(A) \to H_k(A')$ is well defined by equation (1).

Now we compute f_{*k} by taking f_k of representatives of cosets, and we define the group operation of a factor group by applying the group operation of the original group to representatives of cosets. It follows at once from the fact that the action of f_k on $Z_k(A)$ is a homomorphism of $Z_k(A)$ into $Z_k(A')$ that f_{*k} is a homomorphism of $H_k(A)$ into $H_k(A')$. ◆

If the collections of maps f, ∂, and ∂' have the property, given in Theorem 44.4, that the squares are commutative, then f **commutes with** ∂.

After another definition, we shall give a seemingly trivial but very important illustration of Theorem 44.4.

44.5 Definition A chain complex $\langle A', \partial' \rangle$ is a **subcomplex of a chain complex** $\langle A, \partial \rangle$, if, for all k, A'_k is a subgroup of A_k and $\partial'_k(c) = \partial_k(c)$ for every $c \in A'_k$, that is, ∂'_k and ∂_k have the same effect on elements of the subgroup A'_k of A_k. ■

44.6 Example Let A be a chain complex, and let A' be a subcomplex of A. Let i be the collection of injection mappings $i_k : A'_k \to A_k$ given by $i_k(c) = c$ for $c \in A'_k$. It is obvious that i commutes with ∂. Thus we have induced homomorphisms $i_{*k} : H_k(A') \to H_k(A)$. One might naturally suspect that i_{*k} must be an isomorphic mapping of $H_k(A')$ into $H_k(A)$. *This need not be so!* For example, let us consider the 2-sphere S^2 as a subcomplex of the 3-cell E^3. This gives rise to $i_2 : C_2(S^2) \to C_2(E^3)$ and induces

$$i_{*2} : H_2(S^2) \to H_2(E^3).$$

But we have seen that $H_2(S^2) \simeq \mathbb{Z}$, while $H_2(E^3) = 0$. Thus i_{*2} cannot possibly be an isomorphic mapping. ▲

Relative Homology

Suppose that A' is a subcomplex of the chain complex A. The topological situation from which this arises is the consideration of a *simplicial subcomplex* Y (in the obvious sense) of a simplicial complex X. We can then naturally consider $C_k(Y)$ a subgroup of $C_k(X)$, just as in the algebraic situation where we have A'_k a subgroup of A_k. Clearly, we would have

$$\partial_k[C_k(Y)] \leq C_{k-1}(Y).$$

Let us deal now with the algebraic situation and remember that it can be applied to our topological situation at any time.

If A' is a subcomplex of the chain complex A, we can form the collection A/A' of factor groups A_k/A'_k. We claim that A/A' again gives rise to a chain complex in a natural way, and we must exhibit a collection $\bar{\partial}$ of homomorphisms

$$\bar{\partial}_k : (A_k/A'_k) \to (A_{k-1}/A'_{k-1})$$

such that $\bar{\partial}_{k-1}\bar{\partial}_k = 0$. The definition of $\bar{\partial}_k$ to attempt is obvious, namely, define

$$\bar{\partial}_k(c + A'_k) = \partial_k(c) + A'_{k-1}$$

for $c \in A_k$. We have to show three things: that $\bar{\partial}_k$ is well defined, that it is a homomorphism, and that $\bar{\partial}_{k-1}\bar{\partial}_k = 0$.

First, to show that $\bar{\partial}_k$ is well defined, let c_1 also be in $c + A'_k$. Then $(c_1 - c) \in A'_k$, so $\partial_k(c_1 - c) \in A'_{k-1}$. Thus

$$\partial_k(c_1) \in (\partial_k(c) + A'_{k-1})$$

also. This shows that $\bar{\partial}_k$ is well defined.

The equation

$$\begin{aligned}\bar{\partial}_k((c_1 + A'_k) + (c_2 + A'_k)) &= \bar{\partial}_k((c_1 + c_2) + A'_k) \\ &= \partial_k(c_1 + c_2) + A'_{k-1} \\ &= (\partial_k(c_1) + \partial_k(c_2)) + A'_{k-1} \\ &= \bar{\partial}_k(c_1 + A'_k) + \bar{\partial}_k(c_2 + A'_k)\end{aligned}$$

shows that $\bar{\partial}_k$ is a homomorphism.

Finally,

$$\begin{aligned}\bar{\partial}_{k-1}(\bar{\partial}_k(c + A'_k)) &= \bar{\partial}_{k-1}(\partial_k(c) + A'_{k-1}) \\ &= \partial_{k-1}(\partial_k(c)) + A'_{k-2} = 0 + A'_{k-2},\end{aligned}$$

so $\bar{\partial}_{k-1}\bar{\partial}_k = 0$.

The preceding arguments are typical routine computations to the homological algebraist, just as addition and multiplication of integers are routine to you. We gave them in great detail. One has to be a little careful to keep track of *dimension,* i.e., to keep track of subscripts. Actually, the expert in homological algebra usually does not write most of these indices, but he always knows precisely with which group he is working. We gave all the indices so that you could keep track of exactly which groups were under consideration. Let us summarize the above work in a theorem.

44.7 Theorem If A' is a subcomplex of the chain complex A, then the collection A/A' of factor groups A_k/A'_k, together with the collection $\bar{\partial}$ of homomorphisms $\bar{\partial}_k$ defined by

$$\bar{\partial}_k(c + A'_k) = \partial_k(c) + A'_{k-1}$$

for $c \in A_k$, is a chain complex.

Since A/A' is a chain complex, we can then form the homology groups $H_k(A/A')$.

44.8 Definition The homology group $H_k(A/A')$ is the ***k*th relative homology group of *A* modulo *A'*.** ■

In our topological situation where Y is a subcomplex of a simplicial complex X, we shall conform to the usual notation of topologists and denote the kth relative homology group arising from the subcomplex $C(Y)$ of the chain complex $C(X)$ by "$H_k(X, Y)$." All the chains of Y are thus "set equal to 0." Geometrically, this corresponds to shrinking Y to a point.

44.9 Example Let X be the simplicial complex consisting of the edges (excluding the inside) of the triangle in Fig. 44.10, and let Y be the subcomplex consisting of the edge P_2P_3. We have seen that $H_1(X) \simeq H_1(S^1) \simeq \mathbb{Z}$. Shrinking P_2P_3 to a point collapses the rim of the triangle, as shown in Fig. 44.11. The result is still topologically the same as S^1. Thus, we would expect again to have $H_1(X, Y) \simeq \mathbb{Z}$.

Generators for $C_1(X)$ are P_1P_2, P_2P_3, and P_3P_1. Since $P_2P_3 \in C_1(Y)$, we see that generators of $C_1(X)/C_1(Y)$ are

$$P_1P_2 + C_1(Y) \qquad \text{and} \qquad P_3P_1 + C_1(Y).$$

To find $Z_1(X, Y)$ we compute

$$\begin{aligned}\bar{\partial}_1(nP_1P_2 + mP_3P_1 + C_1(Y)) &= \partial_1(nP_1P_2) + \partial_1(mP_3P_1) + C_0(Y)\\ &= n(P_2 - P_1) + m(P_1 - P_3) + C_0(Y)\\ &= (m - n)P_1 + C_0(Y),\end{aligned}$$

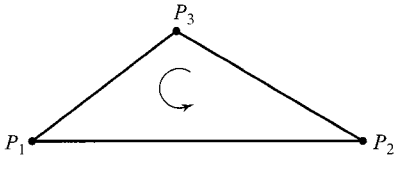
since $P_2, P_3 \in C_0(Y)$. Thus for a cycle, we must have $m = n$, so a generator of $Z_1(X, Y)$ is $(P_1P_2 + P_3P_1) + C_1(Y)$. Since $B_1(X, Y) = 0$, we see that indeed

$$H_1(X, Y) \simeq \mathbb{Z}.$$

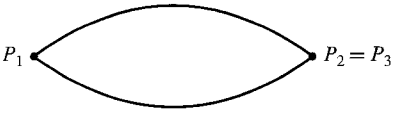
Since $P_1 + C_0(Y)$ generates $Z_0(X, Y)$ and

$$\bar{\partial}_1(P_2P_1 + C_1(Y)) = (P_1 - P_2) + C_0(Y) = P_1 + C_0(Y),$$

we see that $H_0(X, Y) = 0$. This is characteristic of relative homology groups of dimension 0 for connected simplicial complexes. ▲

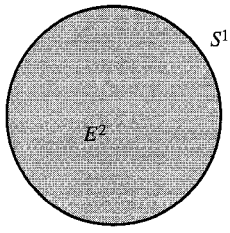


44.10 Figure



44.11 Figure

44.12 Example Let us consider S^1 as a subcomplex (the boundary) of E^2 and compute $H_2(E^2, S^1)$. Remember that E^2 is a circular disk, so S^1 can be indeed thought of as its boundary (see Fig. 44.13). You can demonstrate the shrinking of S^1 to a point by putting a drawstring around the edge of a circular piece of cloth and then drawing the string so that the rim of the circle comes in to one point. The resulting space is then a closed bag or S^2. Thus, while $H_2(E^2) = 0$, since E^2 is a contractible space, we would expect



44.13 Figure

$$H_2(E^2, S^1) \simeq \mathbb{Z}.$$

For purposes of computation, we can regard E^2 topologically as the triangular region of Fig. 44.10 and S^1 as the rim of the triangle. Then $C_2(E^2, S^1)$ is generated by $P_1P_2P_3 + C_2(S^1)$, and

$$\begin{aligned}\bar{\partial}_2\big(P_1P_2P_3 + C_2(S^1)\big) &= \partial_2(P_1P_2P_3) + C_1(S^1)\\ &= (P_2P_3 - P_1P_3 + P_1P_2) + C_1(S^1).\end{aligned}$$

But $(P_2P_3 - P_1P_3 + P_1P_2) \in C_1(S^1)$, so we have

$$\bar{\partial}_2\big(P_1P_2P_3 + C_2(S^1)\big) = 0.$$

Hence $P_1P_2P_3 + C_2(S^1)$ is an element of $Z_2(E^2, S^1)$. Since

$$B_2(E^2, S^1) = 0,$$

we see that

$$H_2(E^2, S^1) \simeq \mathbb{Z},$$

as we expected. ▲

The Exact Homology Sequence of a Pair

We now describe the *exact homology sequence of a pair* and give an application. We shall not carry out all the details of the computations. The computations are routine and straightforward. We shall give all the necessary definitions, and shall let the student supply the details in the exercises.

44.14 Lemma Let A' be a subcomplex of a chain complex A. Let j be the collection of natural homomorphisms $j_k : A_k \to (A_k/A'_k)$. Then

$$j_{k-1}\partial_k = \bar{\partial}_k j_k,$$

that is, j commutes with ∂.

Proof We leave this easy computation to the exercises (see Exercise 12). ◆

44.15 Theorem The map j_k of Lemma 44.14 induces a natural homomorphism

$$j_{*k} : H_k(A) \to H_k(A/A').$$

Proof This is immediate from Lemma 44.14 and Theorem 44.4 ◆

Let A' be a subcomplex of the chain complex A. Let $h \in H_k(A/A')$. Then $h = z + B_k(A/A')$ for $z \in Z_k(A/A')$, and in turn $z = c + A'_k$ for some $c \in A_k$. (Note that we arrive at c from h by two successive choices of representatives.) Now $\bar{\partial}_k(z) = 0$, which implies that $\partial_k(c) \in A'_{k-1}$. This, together with $\partial_{k-1}\partial_k = 0$, gives us $\partial_k(c) \in Z_{k-1}(A')$. Define

$$\partial_{*k} : H_k(A/A') \to H_{k-1}(A')$$

by

$$\partial_{*k}(h) = \partial_k(c) + B_{k-1}(A').$$

This definition of ∂_{*k} looks very complicated. Think of it as follows. Start with an element of $H_k(A/A')$. Now such an element is represented by a relative k-cycle modulo A'. To say it is a relative k-cycle modulo A' is to say that its boundary is in A'_{k-1}. Since its boundary is in A'_{k-1} and is a boundary of something in A_k, this boundary must be a $(k-1)$-cycle in A'_{k-1}. Thus starting with $h \in H_k(A/A')$, we have arrived at a $(k-1)$-cycle representing a homology class in $H_{k-1}(A')$.

44.16 Lemma The map $\partial_{*k} : H_k(A/A') \to H_{k-1}(A')$, which we have just defined, is well defined, and is a homomorphism of $H_k(A/A')$ into $H_{k-1}(A')$.

Proof We leave this proof to the exercises (see Exercise 13). ◆

Let i_{*k} be the map of Example 44.6. We now can construct the following diagram.

$$\begin{aligned} \cdots \xrightarrow{\partial_{*k+1}} H_k(A') &\xrightarrow{i_{*k}} H_k(A) \xrightarrow{j_{*k}} H_k(A/A') \\ \xrightarrow{\partial_{*k}} H_{k-1}(A') &\xrightarrow{i_{*k-1}} H_{k-1}(A) \xrightarrow{j_{*k-1}} H_{k-1}(A/A') \xrightarrow{\partial_{*k-1}} \cdots \end{aligned} \tag{1}$$

44.17 Lemma The groups in diagram (1), together with the given maps, form a chain complex.

Proof You need only check that a sequence of two consecutive maps always gives 0. We leave this for the exercises (see Exercise 14). ◆

Since diagram (1) gives a chain complex, we could (horrors!) ask for the homology groups of this chain complex. We have been aiming at this question, the answer to which is actually quite easy. *All the homology groups of this chain complex are* 0. You may think that such a chain complex is uninteresting. Far from it. Such a chain complex even has a special name.

44.18 Definition A sequence of groups A_k and homomorphisms ∂_k forming a chain complex is an **exact sequence** if all the homology groups of the chain complex are 0, that is, if for all k we have that the image under ∂_k is equal to the kernel of ∂_{k-1}.

Exact sequences are of great importance in topology. We shall give some elementary properties of them in the exercises. ■

44.19 Theorem The groups and maps of the chain complex in diagram (1) form an exact sequence.

Proof We leave this proof to the exercises (see Exercise 15). ◆

44.20 Definition The exact sequence in diagram (1) is the **exact homology sequence of the pair** (A, A'). ■

44.21 Example Let us now give an application of Theorem 44.19 to topology. We have stated without proof that $H_n(S^n) \simeq \mathbb{Z}$ and $H_0(S^n) \simeq \mathbb{Z}$, but that $H_k(S^n) = 0$ for $k \neq 0, n$. We have also stated without proof that $H_k(E^n) = 0$ for $k \neq 0$, since E^n is contractible. Let us assume the result for E^n and now *derive* from this the result for S^n.

We can view S^n as a subcomplex of the simplicial complex E^{n+1}. For example, E^{n+1} is topologically equivalent to an $(n+1)$-simplex, and S^n is topologically equivalent to its boundary. Let us form the exact homology sequence of the pair (E^{n+1}, S^n). We have

$$\begin{aligned}
&\underbrace{H_{n+1}(S^n)}_{=0} \xrightarrow{i_{*n+1}} \underbrace{H_{n+1}(E^{n+1})}_{=0} \xrightarrow{j_{*n+1}} \underbrace{H_{n+1}(E^{n+1}, S^n)}_{\simeq \mathbb{Z}} \xrightarrow{\partial_{*n+1}} \\
&\underbrace{H_n(S^n)}_{=?} \xrightarrow{i_{*n}} \underbrace{H_n(E^{n+1})}_{=0} \xrightarrow{j_{*n}} \underbrace{H_n(E^{n+1}, S^n)}_{=0} \xrightarrow{\partial_{*n}} \cdots \xrightarrow{j_{*k+1}} \\
&\underbrace{H_{k+1}(E^{n+1}, S^n)}_{=0} \xrightarrow{\partial_{*k+1}} \underbrace{H_k(S^n)}_{=?} \xrightarrow{i_{*k}} \underbrace{H_k(E^{n+1})}_{=0} \xrightarrow{j_{*k}} \cdots
\end{aligned} \tag{2}$$

for $1 \leq k < n$. The fact that E^{n+1} is contractible gives $H_k(E^{n+1}) = 0$ for $k \geq 1$. We have indicated this on diagram (2). Viewing E^{n+1} as an $(n+1)$-simplex and S^n as its boundary, we see that $C_k(E^{n+1}) \leq C_k(S^n)$ for $k \leq n$. Therefore $H_k(E^{n+1}, S^n) = 0$ for $k \leq n$. We also indicated this on diagram (2). Just as in Example 44.12, one sees that $H_{n+1}(E^{n+1}, S^n) \simeq \mathbb{Z}$, with a generating homology class containing as representative

$$P_1 P_2 \cdots P_{n+2} + C_{n+1}(S^n).$$

For $1 \leq k < n$, the exact sequence in the last row of diagram (2) tells us that $H_k(S^n) = 0$, for from $H_k(E^{n+1}) = 0$, we see that

$$(\text{kernel } i_{*k}) = H_k(S^n).$$

But from $H_{k+1}(E^{n+1}, S^n) = 0$, we see that (image ∂_{*k+1}) $= 0$. From exactness, (kernel i_{*k}) = (image ∂_{*k+1}), so $H_k(S^n) = 0$ for $1 \leq k < n$.

The following chain of reasoning leads to $H_n(S^n) \simeq \mathbb{Z}$. Refer to diagram (2) above.

1. Since $H_{n+1}(E^{n+1}) = 0$, we have (image j_{*n+1}) $= 0$.
2. Hence (kernel ∂_{*n+1}) = (image j_{*n+1}) $= 0$ by exactness, that is, ∂_{*n+1} is an isomorphic mapping.
3. Therefore (image ∂_{*n+1}) $\simeq \mathbb{Z}$.
4. Since $H_n(E^{n+1}) = 0$, we have (kernel i_{*n}) $= H_n(S^n)$.
5. By exactness, (image ∂_{*n+1}) = (kernel i_{*n}), so $H_n(S^n) \simeq \mathbb{Z}$.

Thus we see that $H_n(S^n) \simeq \mathbb{Z}$ and $H_k(S^n) = 0$ for $1 \leq k < n$.

Since S^n is connected, $H_0(S^n) \simeq \mathbb{Z}$. This fact could also be deduced from the exact sequence

$$\underbrace{H_1(E^{n+1}, S^n)}_{=0} \xrightarrow{\partial_{*1}} H_0(S^n) \xrightarrow{i_{*0}} \underbrace{H_0(E^{n+1})}_{\simeq \mathbb{Z}} \xrightarrow{j_{*0}} \underbrace{H_0(E^{n+1}, S^n)}_{=0}.$$

▲

EXERCISES 44

Suggested Exercises

1. Let A and B be additive groups, and suppose that the sequence

$$0 \to A \xrightarrow{f} B \to 0$$

is exact. Show that $A \simeq B$.

2. Let A, B, and C be additive groups and suppose that the sequence

$$0 \to A \xrightarrow{i} B \xrightarrow{j} C \to 0$$

is exact. Show that

a. j maps B onto C

b. i is an isomorphism of A into B

c. C is isomorphic to $B/i[A]$

3. Let A, B, C, and D be additive groups and let

$$A \xrightarrow{i} B \xrightarrow{j} C \xrightarrow{k} D$$

be an exact sequence. Show that the following three conditions are equivalent:

a. i is onto B

b. j maps all of B onto 0

c. k is a one-to-one map

4. Show that if

$$A \xrightarrow{g} B \xrightarrow{h} C \xrightarrow{i} D \xrightarrow{j} E \xrightarrow{k} F$$

is an exact sequence of additive groups, then the following are equivalent:

a. h and j both map everything onto 0

b. i is an isomorphism of C onto D

c. g is onto B and k is one to one

More Exercises

5. Theorem 44.4 and Theorem 44.7 are closely connected with Exercise 39 of Section 14. Show the connection.

6. In a computation analogous to Examples 44.9 and 44.12 of the text, find the relative homology groups $H_n(X, a)$ for the torus X with subcomplex a, as shown in Figs. 42.13 and Fig. 42.14. (Since we can regard these relative homology groups as the homology groups of the space obtained from X by shrinking a to a point, these should be the homology groups of the *pinched torus*.)

7. For the simplicial complex X and subcomplex a of Exercise 6, form the exact homology sequence of the pair (X, a) and verify by direct computation that this sequence is exact.
8. Repeat Exercise 6 with X the Klein bottle of Fig. 43.2 and Fig. 43.4. (This should give the homology groups of the *pinched Klein bottle*.)
9. For the simplicial complex X and subcomplex a of Exercise 8, form the exact homology sequence of the pair (X, a) and verify by direct computation that this sequence is exact.
10. Find the relative homology groups $H_n(X, Y)$, where X is the annular region of Fig. 42.11 and Y is the subcomplex consisting of the two boundary circles.
11. For the simplicial complex X and subcomplex Y of Exercise 10, form the exact homology sequence of the pair (X, Y) and verify by direct computation that this sequence is exact.
12. Prove Lemma 44.14
13. Prove Lemma 44.16
14. Prove Lemma 44.17
15. Prove Theorem 44.19 by means of the following steps.
    a. Show (image i_{*k}) $\subseteq$ (kernel j_{*k}).
    b. Show (kernel j_{*k}) $\subseteq$ (image i_{*k}).
    c. Show (image j_{*k}) $\subseteq$ (kernel ∂_{*k}).
    d. Show (kernel ∂_{*k}) $\subseteq$ (image j_{*k}).
    e. Show (image ∂_{*k}) $\subseteq$ (kernel i_{*k-1}).
    f. Show (kernel i_{*k-1}) $\subseteq$ (image ∂_{*k}).
16. Let $\langle A, \partial \rangle$ and $\langle A', \partial' \rangle$ be chain complexes, and let f and g be collections of homomorphisms $f_k : A_k \to A'_k$ and $g_k : A_k \to A'_k$ such that both f and g commute with ∂. An **algebraic homotopy** between f and g is a collection D of homomorphisms $D_k : A_k \to A'_{k+1}$ such that for all $c \in A_k$, we have

$$f_k(c) - g_k(c) = \partial'_{k+1}(D_k(c)) + D_{k-1}(\partial_k(c)).$$

(One abbreviates this condition by $f - g = \partial D + D\partial$.) Show that if there exists an algebraic homotopy between f and g, that is, if f and g are **homotopic,** then f_{*k} and g_{*k} are the same homomorphism of $H_k(A)$ into $H_k(A')$.